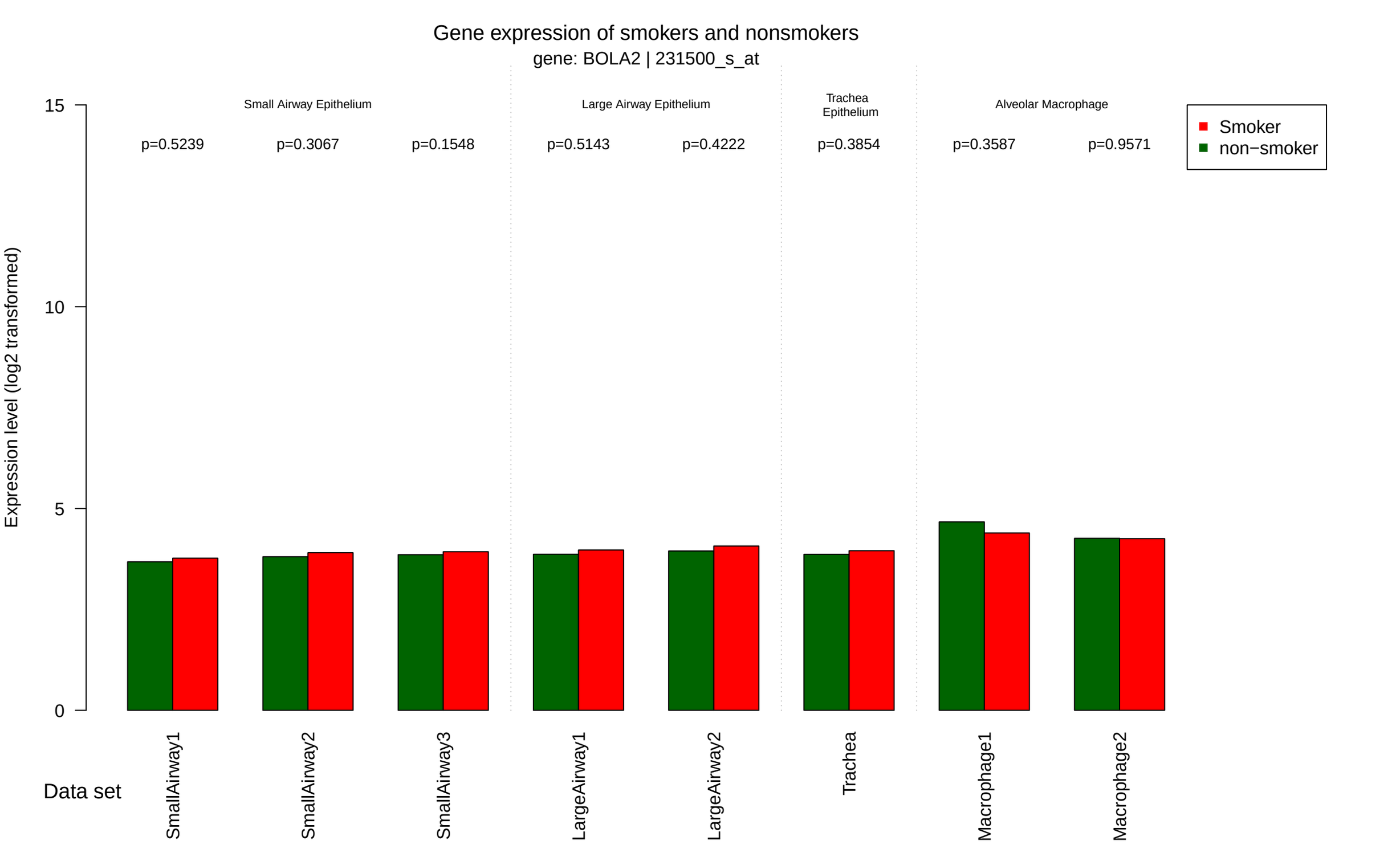
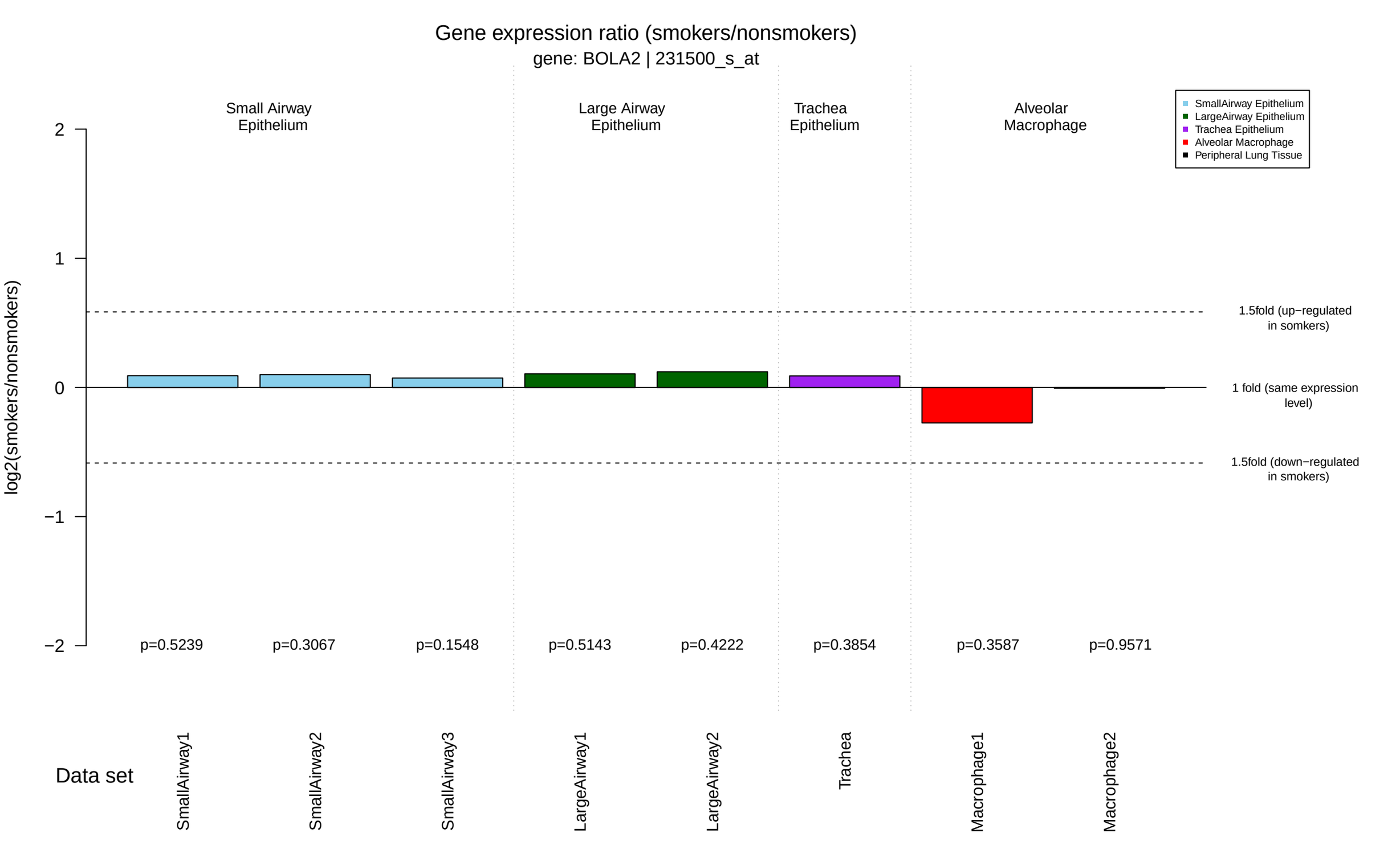
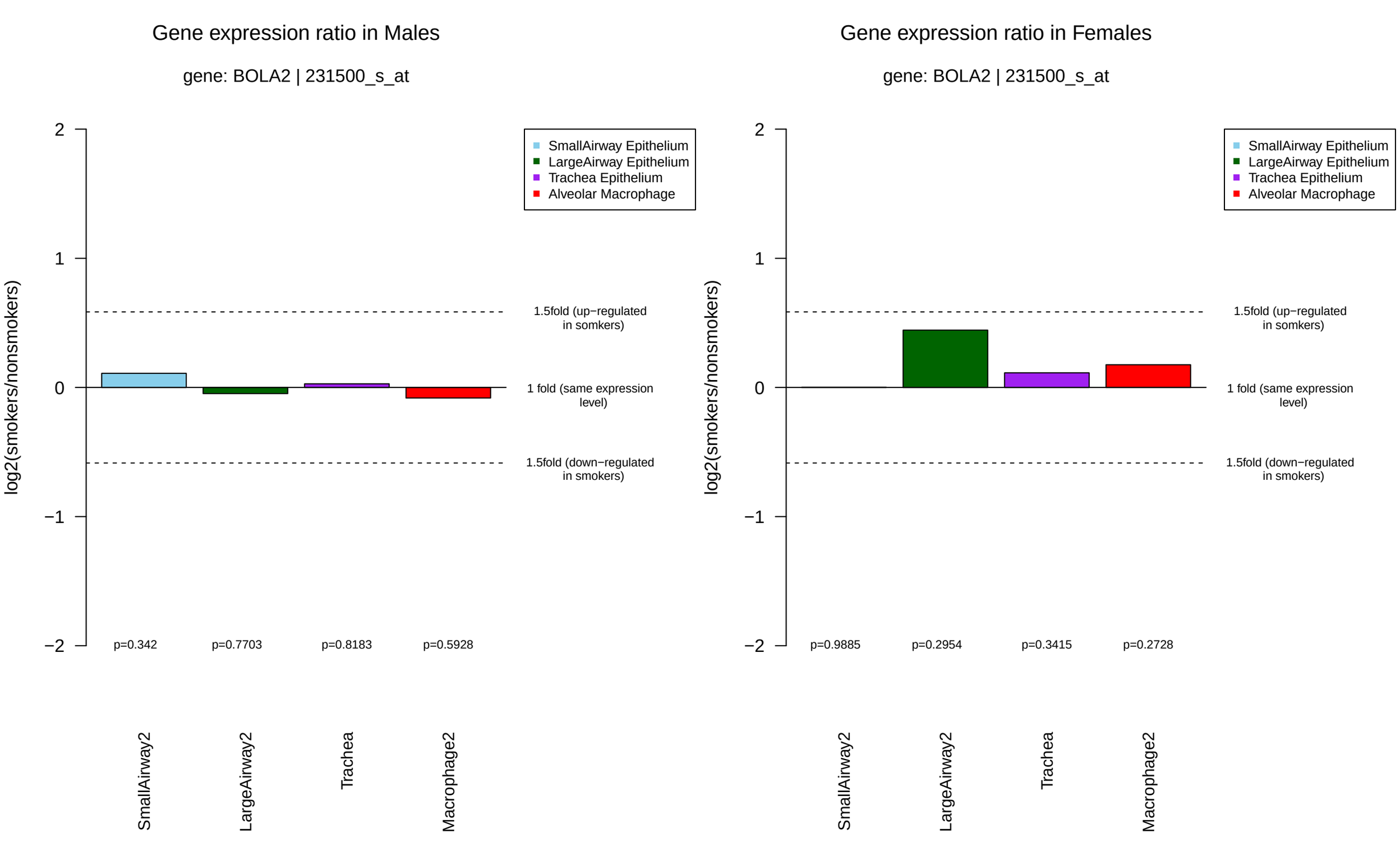
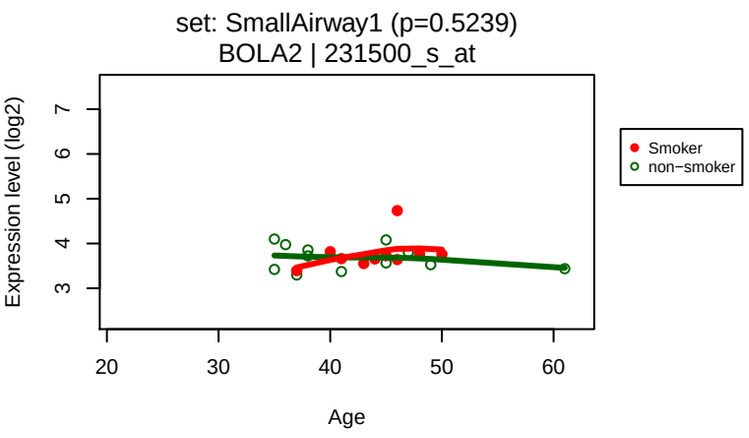
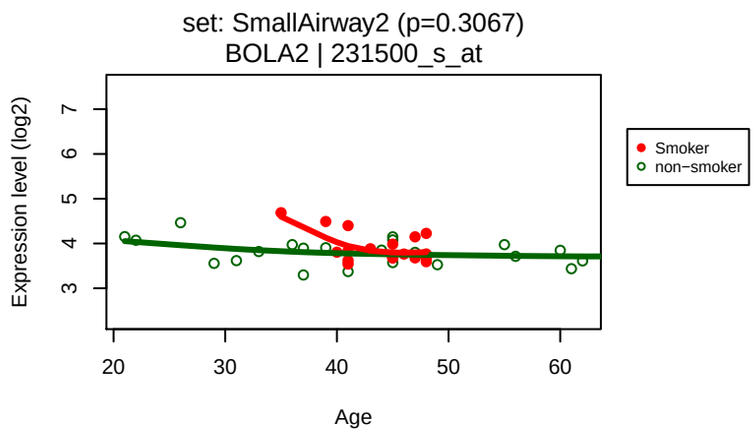
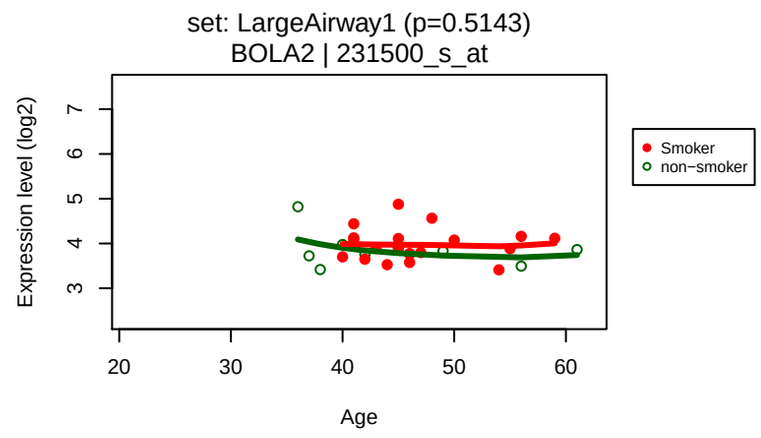

Gene expression of smokers and nonsmokers
gene: BOLA2 | 231500_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5239
p=0.3067
p=0.1548
p=0.5143
p=0.4222
p=0.3854
p=0.3587
p=0.9571
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

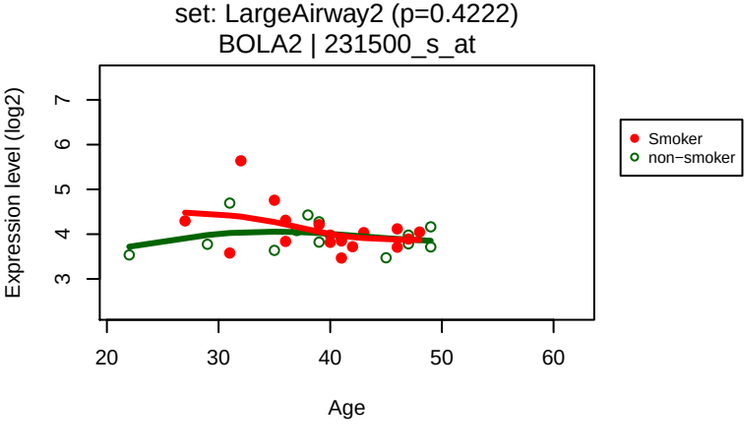

Gene expression ratio (smokers/nonsmokers)
gene: BOLA2 | 231500_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5239
p=0.3067
p=0.1548
p=0.5143
p=0.4222
p=0.3854
p=0.3587
p=0.9571
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

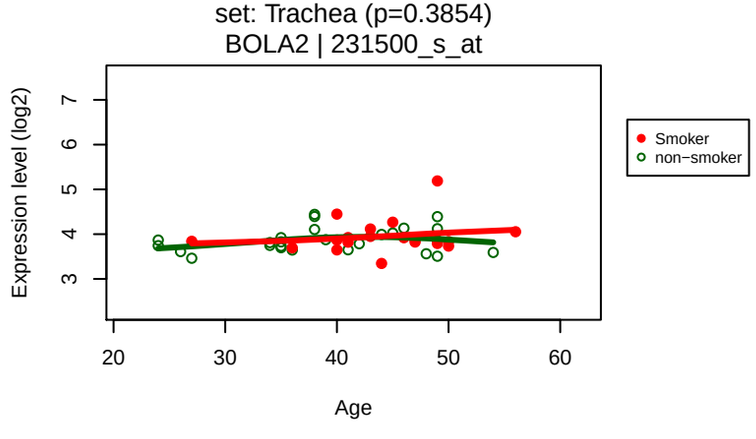

Gene expression ratio in Males
gene: BOLA2 | 231500_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.342
p=0.7703
p=0.8183
p=0.5928
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BOLA2 | 231500_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9885
p=0.2954
p=0.3415
p=0.2728
SmallAirway2
LargeAirway2
Trachea
Macrophage2

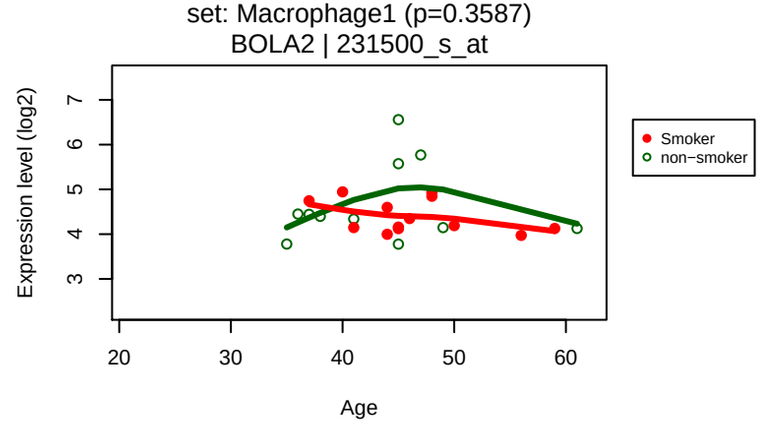

set: SmallAirway1 (p=0.5239)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker

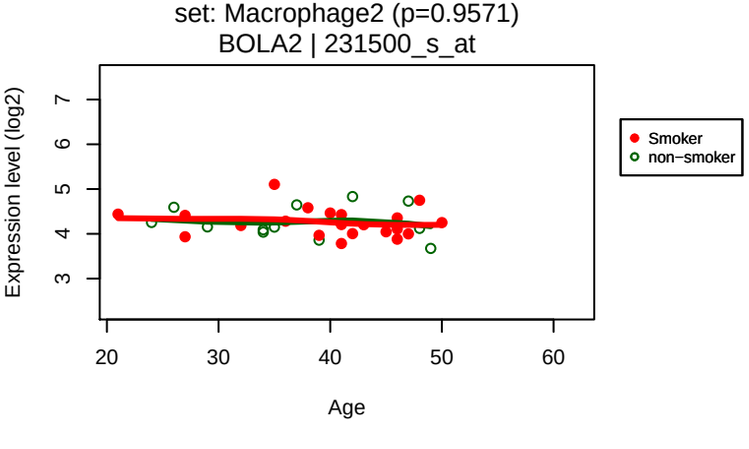

set: SmallAirway2 (p=0.3067)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker

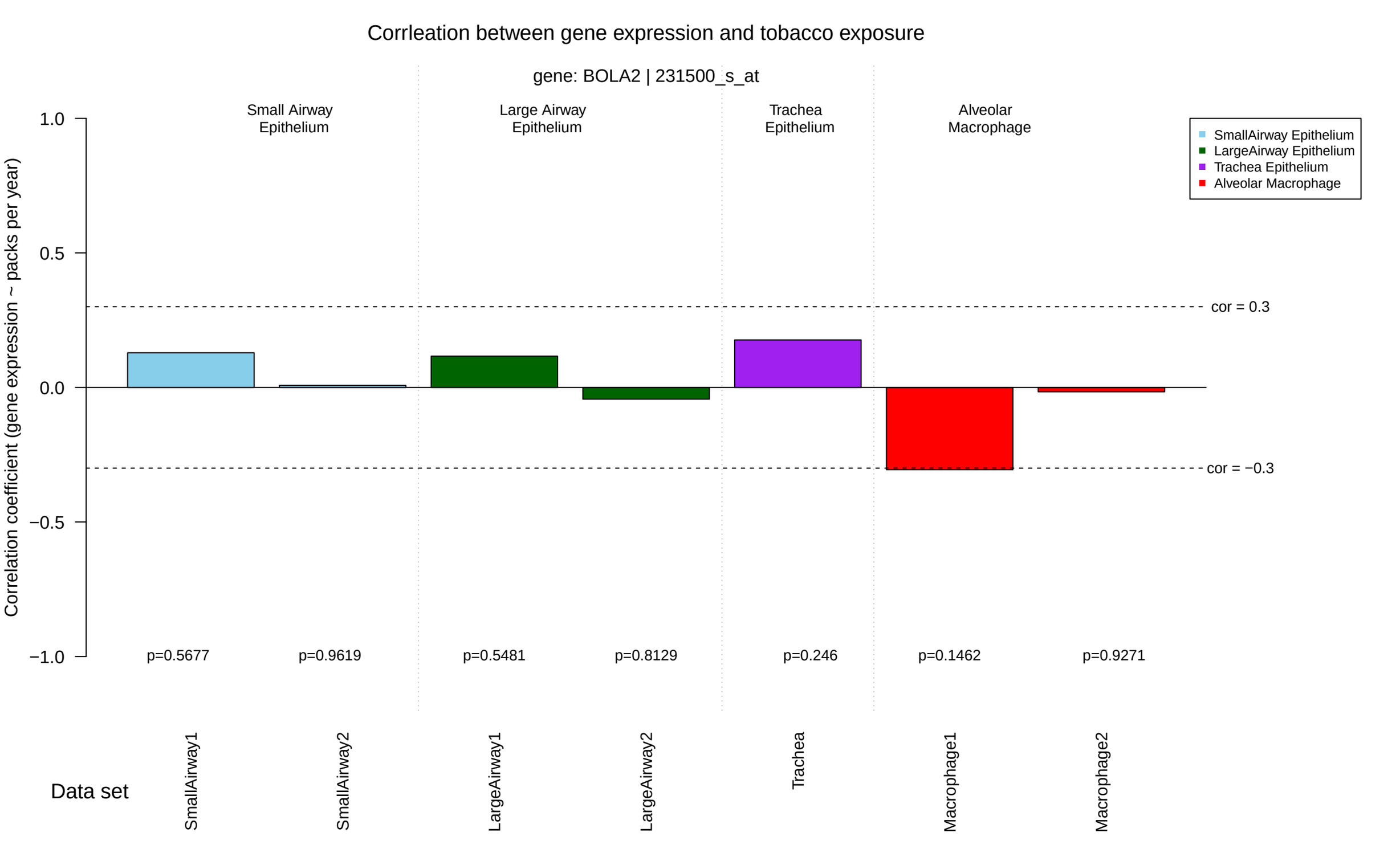

set: LargeAirway1 (p=0.5143)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4222)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3854)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3587)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9571)
BOLA2 | 231500_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BOLA2 | 231500_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5677
p=0.9619
p=0.5481
p=0.8129
p=0.246
p=0.1462
p=0.9271
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2